

HAGI, Japan

WMO: 477540

Lat: 34.4100N

Lon: 131.4061E

Elev: 12

StdP: 14.69

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 32.0	(c) 33.5	(d) 18.1	(e) 13.7	(f) 38.1	(g) 20.4	(h) 15.3	(i) 39.3	(j) 23.0	(k) 39.7	(l) 20.3	(m) 40.4	(n) 9.0	(o) 170	(p) 0.390

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 12.3	(c) 91.8	(d) 78.0	(e) 89.7	(f) 78.0	(g) 87.7	(h) 77.8	(i) 80.1	(j) 87.7	(k) 79.2	(l) 86.9	(m) 78.3	(n) 85.8	(o) 9.1	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.1	146.0	84.7	77.1	140.9	83.9	76.1	136.5	83.1	43.8	88.0	42.8	87.1	41.9	86.1	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 18.8	(b) 16.0	(c) 13.6	(e) 28.9	(f) 95.8	(g) 2.5	(h) 1.9	(i) 27.1	(j) 97.2	(k) 25.6	(l) 98.3	(m) 24.1	(n) 99.4	(o) 22.3	(p) 100.8		
(a) 18.8	(b) 16.0	(c) 13.6	(e) 26.2	(f) 81.5	(g) 2.6	(h) 1.1	(i) 24.3	(j) 82.3	(k) 22.7	(l) 83.0	(m) 21.2	(n) 83.6	(o) 19.3	(p) 84.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.0	42.6	43.6	48.5	56.8	65.1	71.6	79.7	81.8	74.3	64.7	55.6	46.8
	DBStd	14.24	4.26	5.16	5.54	5.63	4.81	4.38	4.45	3.76	4.95	4.79	5.42	4.95
	HDD50	667	235	190	97	10	0	0	0	0	0	0	11	124
	HDD65	3031	696	598	512	252	59	2	0	0	0	62	286	564
	CDD50	4689	4	12	49	214	468	649	920	985	729	456	178	25
	CDD65	1577	0	0	0	6	61	200	455	520	279	53	3	0
	CDH74	12927	0	0	1	29	241	883	4262	5494	1836	178	3	0
	CDH80	4193	0	0	0	4	27	162	1454	2105	424	17	0	0
Wind	WSAvg	6.6	7.6	7.4	7.2	6.9	6.6	5.4	5.8	6.0	5.5	6.0	6.8	7.6
Precipitation	PrecAvg	68.7	3.5	3.2	4.9	5.5	6.5	9.1	11.1	7.0	7.1	3.8	3.7	3.3
	PrecMax	95.2	7.2	4.9	9.5	11.3	13.7	18.3	25.6	21.6	14.9	8.7	8.7	6.5
	PrecMin	47.8	0.9	0.8	2.4	2.0	1.9	2.0	1.7	1.2	2.1	0.6	1.1	0.6
	PrecStd	13.0	1.5	1.1	2.0	2.4	2.8	4.2	7.0	4.5	3.6	2.3	2.0	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.8	65.2	70.2	77.5	83.3	88.9	93.5	95.1	90.9	82.3	73.6	63.9
		MCWB	53.7	58.6	56.2	61.2	66.1	75.3	78.2	78.5	77.1	71.6	63.4	56.4
	2%	DB	55.3	59.6	65.0	72.8	79.5	84.2	91.1	92.6	87.3	78.2	69.9	60.5
		MCWB	49.1	52.8	55.5	60.9	64.6	72.8	78.0	77.9	76.5	69.9	61.6	54.0
	5%	DB	52.9	55.9	61.8	69.6	76.5	81.2	89.0	90.6	84.5	75.6	67.5	58.0
		MCWB	46.8	49.9	54.1	59.4	64.2	72.2	77.9	78.2	75.5	67.2	60.3	51.5
	10%	DB	50.5	52.8	58.8	66.8	73.9	78.7	87.0	88.6	82.1	73.4	65.2	55.7
		MCWB	44.3	46.9	52.2	58.0	63.9	71.4	77.7	78.3	74.2	65.2	58.4	49.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.6	59.0	60.5	67.5	71.5	77.4	81.0	81.7	79.3	74.8	66.3	59.5
		MCDB	58.2	63.9	64.6	72.7	77.1	85.3	88.2	88.6	86.8	80.4	69.7	61.7
	2%	WB	51.1	54.6	57.6	63.7	68.9	75.6	79.9	80.6	77.6	72.0	64.0	55.4
		MCDB	54.1	58.2	62.5	69.3	74.0	81.5	87.3	88.3	84.5	76.5	68.4	58.7
	5%	WB	47.6	50.8	55.4	61.2	67.0	74.3	79.0	79.7	76.5	69.3	61.8	52.9
		MCDB	51.6	54.6	60.3	66.9	72.8	79.3	86.6	87.4	82.7	73.8	66.2	56.8
	10%	WB	45.1	47.9	52.9	59.5	65.5	72.6	78.1	78.9	75.3	66.8	59.4	50.4
		MCDB	49.5	52.2	58.0	64.7	71.8	77.1	85.3	86.6	81.0	71.4	64.1	54.9
Mean Daily Temperature Range	5% DB	MDBR	11.4	12.8	14.9	15.6	14.9	11.5	11.1	12.3	12.4	14.6	14.0	12.3
		MCDBR	15.8	17.9	19.7	20.1	19.1	14.7	13.9	14.5	14.0	16.1	16.1	15.8
		MCWBR	10.9	12.6	12.7	11.4	9.2	6.8	5.4	5.3	6.4	8.5	9.8	10.7
	5% WB	MCDBR	14.1	14.8	16.0	15.1	14.0	11.4	12.7	13.3	12.0	12.9	13.3	13.6
		MCWBR	11.4	12.2	12.2	9.9	8.5	6.1	5.6	5.9	6.0	7.7	9.3	10.8
Clear-Sky Solar Irradiance	taub	0.439	0.483	0.551	0.577	0.576	0.592	0.591	0.549	0.509	0.464	0.441	0.431	
	taud	2.017	1.941	1.792	1.784	1.842	1.889	1.938	2.061	2.112	2.124	2.102	2.051	
	Ebn at Noon	231	235	230	233	235	230	229	237	240	239	230	225	
	Edh at Noon	43	52	66	69	66	63	60	52	47	43	40	40	
All-Sky Solar Radiation	RadAvg	621	867	1209	1513	1680	1475	1541	1618	1274	1072	772	580	
	RadStd	56	80	96	125	187	141	234	179	123	109	62	39	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page